Name: _____ Period: _____

Earth and Space Science

Where Does the Carbon Go?

Learning Intention
Today we will be learning how carbon moves between Earth's spheres and where it is stored so we can explain why the carbon dioxide released by volcanoes has not built up in the atmosphere over millions of years.
Success Criteria
● I can describe how carbon moves between the atmosphere, biosphere, hydrosphere, and geosphere. ● I can compare the fast and slow carbon cycles using timescales and examples. ● I can use numbers from a carbon cycle model to compare how much carbon different reservoirs store.

Word Bank
Reservoir: a place where carbon is stored (atmosphere, ocean, plants, soil, rock) **Biomass:** the material that makes up living things, like wood and leaves **Combustion:** burning **Carbonic acid:** a weak acid that forms when CO_2 dissolves in water **Weathering:** the breakdown of rock at Earth's surface **Sediment:** pieces of rock, shells, and other material that settle out of water or air **Calcium carbonate ($CaCO_3$):** the material in shells, coral, and limestone

Think first: Volcanoes have been adding CO_2 to the atmosphere for billions of years. Why hasn't it all built up? Write your prediction. You will come back to this in Part B.

_____

_____

Part A: The Fast Carbon Cycle

Video: Model Behavior: Visualizing Global Carbon Dioxide (NASA, 2024, 1:51) https://tinyurl.com/3r5swvtd

This video shows a computer model of carbon dioxide moving through Earth's atmosphere. Answer as you watch. You will see it twice.

1. Name one region of the world where you see large amounts of CO_2 being released. _____

2. Besides power plants and cars, what else releases CO_2 over Africa and South America? _____

3. Watch the CO_2 coming from one place, such as the United States or China. Does it stay over that place? Describe where it goes.

_____

_____

4. Is CO_2 spread evenly around the world, or is it more concentrated in some places? Use something you saw in the video to support your answer.

Reading: Carbon Cycle (Understanding Global Change, UC Berkeley) https://tinyurl.com/2nyrwdfu
Read the opening paragraph and the first paragraph under **"What is the carbon cycle?"**

5. Which sphere contains most of Earth's carbon? _____

6. Name two rocks or fuels that contain carbon. _____ and _____

7. In the fast carbon cycle, carbon moves between the ocean, atmosphere, soil, and living things over time scales of _____ to _____.

8. How do plants remove CO_2 from the atmosphere, and what happens to those carbon atoms?

_____

9. List three ways carbon returns to the atmosphere from living things. _____, _____, _____

Part B: The Slow Carbon Cycle

Read paragraphs 2 and 3 under **"What is the carbon cycle?"** Stop when you reach the paragraph about human activity. All answers are in the reading.

10. In the slow carbon cycle, carbon moves between rocks, the ocean, and the atmosphere over _____ to _____ of years.

11. Number the steps in order from 1 to 6.

____ Shells are buried and the layers are compressed and cemented into limestone.

____ Weathering of rocks removes CO_2 from the atmosphere.

____ Carbon becomes part of calcium carbonate shells made by ocean organisms.

____ Heat and pressure melt the carbon-bearing rock.

____ Sediments are eroded from the land and carried into the ocean.

____ Carbon returns to the atmosphere as CO_2 through volcanism.

12. Besides melted rock, where else does volcanic CO_2 come from? _____

13. Land plants mainly form _____. Ocean plankton mainly form _____ and _____.

14. How long can carbon stay buried in fossil fuels? _____

15. Go back to your Think first prediction. Which process removes volcanic CO_2 from the atmosphere, and where does that carbon end up?

_____

_____

Part C: Sort the Processes

For each process, write the sphere carbon moves **from** and **to** (atmosphere, biosphere, hydrosphere, or geosphere). Then circle **Fast** or **Slow**.

Process	From	To	Fast or Slow
Photosynthesis			Fast / Slow
Respiration			Fast / Slow
Wildfire			Fast / Slow
Ocean absorbs CO_2			Fast / Slow
Weathering of rock by carbonic acid			Fast / Slow
Shells buried and turned into limestone			Fast / Slow
Fossil fuel formation			Fast / Slow
Volcanism			Fast / Slow

Part D: Exit Ticket

The table shows how much carbon is stored in some of Earth's reservoirs.

Reservoir	Carbon Stored (gigatons)
Atmosphere	750
Vegetation	610
Soils	1,580
Surface ocean	1,020
Intermediate and deep ocean	38,100

Source: NYS Earth and Space Sciences Regents, August 2026

16. Which reservoir in the table stores the most carbon? _____

17. About how many times more carbon is stored in the intermediate and deep ocean than in the atmosphere? Show your work.

_____

18. Carlsbad Caverns National Park is carved into limestone that formed from an ancient reef. Explain how carbon that was once in the atmosphere ended up in this rock. Use at least two processes from today.

_____

_____

_____

19. Which table correctly identifies how carbon moves during each process and how fast the process is?

(1)

Process	Carbon Moves	Speed
Photosynthesis	Atmosphere → Biosphere	Fast
Volcanism	Geosphere → Atmosphere	Fast

(2)

Process	Carbon Moves	Speed
Photosynthesis	Biosphere → Atmosphere	Fast
Volcanism	Geosphere → Atmosphere	Slow

(3)

Process	Carbon Moves	Speed
Photosynthesis	Atmosphere → Biosphere	Fast
Volcanism	Geosphere → Atmosphere	Slow

(4)

Process	Carbon Moves	Speed
Photosynthesis	Atmosphere → Biosphere	Slow
Volcanism	Atmosphere → Geosphere	Slow